# Kevin Pei

New York, NY | kevinpei.dev | kevinxpei@gmail.com | linkedin.com/in/kevin-pei | github.com/ktpei

## Education

**New York University** — New York, NY
*B.S. in Computer Science, Data Science Minor | GPA:3.83/4.00* — *Expected Dec. 2027*

## Experience

**Software Engineer Intern** — May 2026 – Present
*Korr Insurtech* — *New York City*

- Owned the implementation of automatic and periodic inflation adjustments for Prudential's Long Term Care insurance lines. Touched 350k+ policies live in production.
- Led the migration of a legacy document template storage system by designing a dynamic entity, developing OpenSearch queries on AWS Lambda, and reconciled entries on DynamoDB. Cut query times by over 70%
- Consolidated 60+ HTTP Lambda functions into a single Express Lambda behind API Gateway, reducing cold-start surfaces from 60 to 1, provisioned memory by 93%, and monthly infrastructure cost by 29%
- Worked directly with clients across 5+ features to identify goals, design solutions, gather feeback, and ship.

**Server Development Lead** — Dec. 2025 – Present
*NYU High Speed Research Lab* — *New York City*

- Extend Corelink's open source JavaScript/Python libraries for data-agnostic, real-time transport applications.
- Reduced streaming jitter by 20% and added support for 4+ protocols, enabling browser-native streaming
- Built plugin framework to offload real-time processing to containerized workloads; specified CLI workflows for composing research pipelines and managing plugin lifecycles.

**Software Engineer Intern** — Jun. 2025 – Aug. 2025
*INF.ai* — *New York City*

- Scaled a securities-exam prep platform to support 500+ concurrent users by optimizing query and request handling
- Implemented role-based access across test modules, admin dashboards, and performance analytics for instructors
- Built timed-exam state management with Valtio, enforcing exam progress and sequential navigation

## Projects

**PersonaLab** (1st at NYU Buildathon) | *TypeScript, Next.js, BullMQ, PostgreSQL, Docker* — Feb. 2026

- Built an automated UX-testing platform that deploys browser agents to simulate user personas across product flows
- Developed an asynchronous BullMQ + Redis pipeline supporting 12+ concurrent browser simulation
- Engineered agentic loop with retry logic, state recovery, and stall detection to ensure reliable agent behavior across multi-step e-commerce flows and browser tasks.

**AgentOS** (1st at Claude Hackathon Columbia) | *Python, FastAPI, Docker, Next.js, Supabase, OAuth* — Apr. 2026

- Built a Fiverr-style platform for configuring and deploying specialized OpenClaw agents
- Developed a FastAPI and Docker orchestration layer that provisions isolated, role-configured agents with persistent memory and scoped OAuth integrations
- Designed a private Docker bridge network for task dispatch between platform and worker containers

**Compco (Co-founder)** | *TypeScript, React, Node.js, Figma, AliCloud* — Apr. 2023 – Aug. 2024

- Launched a Looking-For-Group platform for community events across Shanghai, 200+ monthly active
- Built and deployed backend services, authentication, and account management on Alibaba Cloud

## Technical Skills

**Languages**: Python, JavaScript/TypeScript, SQL
**Frameworks**: React, React Native, Next.js, Node.js, FastAPI
**Cloud & Infra**: AWS, GCP, DynamoDB, PostgreSQL, S3, SQS, Redis, OpenSearch, ElastiCache, Docker
**Developer Tools**: Git, Claude Code, Cursor, Supabase, Figma
**Work Authorization**: United States - No sponsorship required